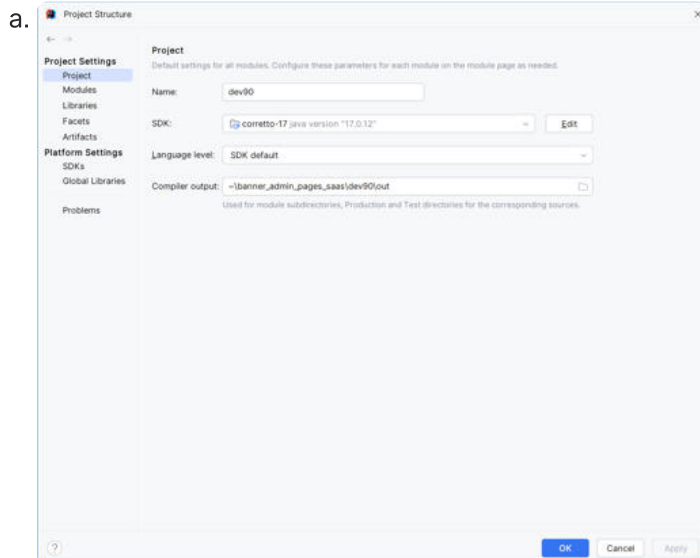


How to Install Banner Admin in IntelliJ

This guide is tested in IntelliJ IDEA 2024.1.4 Ultimate Edition with the updated UI enabled

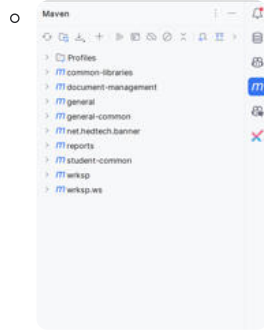
1. Create a new folder for your admin repositories.
2. Clone your desired [Admin repositories](#) into that folder. You can use the [clone-repos.sh](#) script to clones the dev branch for most of the Admin repositories that we use. Make sure you are on the dev or dev_saas branch for all of your repositories or else some build steps may not work!
3. Open IntelliJ, and open that folder that contains all of your repositories (You do not need a dev90workspace folder). Make sure to select “Open Project” and not “New Project”.
4. Set your Java version to Java 17 in the Project Structure menu.



5. Update your Maven settings.xml file as shown in the [JDK17 / Spring 6 Migration Guide](#) to allow you to use HTTP dependencies.
6. In your project view, right click the `.pom` files of the projects you would like to add and click add as maven project.
 - At the bare minimum, you need the following `pom.xml` files
 - i. base\net.hedtech.banner\pom.xml
 - ii. common\common-libraries
 - iii. common\document-management\pom.xml
 - iv. common\general-common\pom.xml
 - v. common\student-common\pom.xml
 - vi. general\workspace-forms\general\pom.xml
 - vii. reports\pom.xml

viii. workspace\webapp-workspace\pom.xml

ix. build\webapp-services\pom.xml



7. Reload your Maven project by going to the clicking on the Maven tab on the right side of your screen. If you do not see it, press ctrl (command on Mac)+shift+a and type in Maven to open the Maven menu. Then, click on the refresh button.

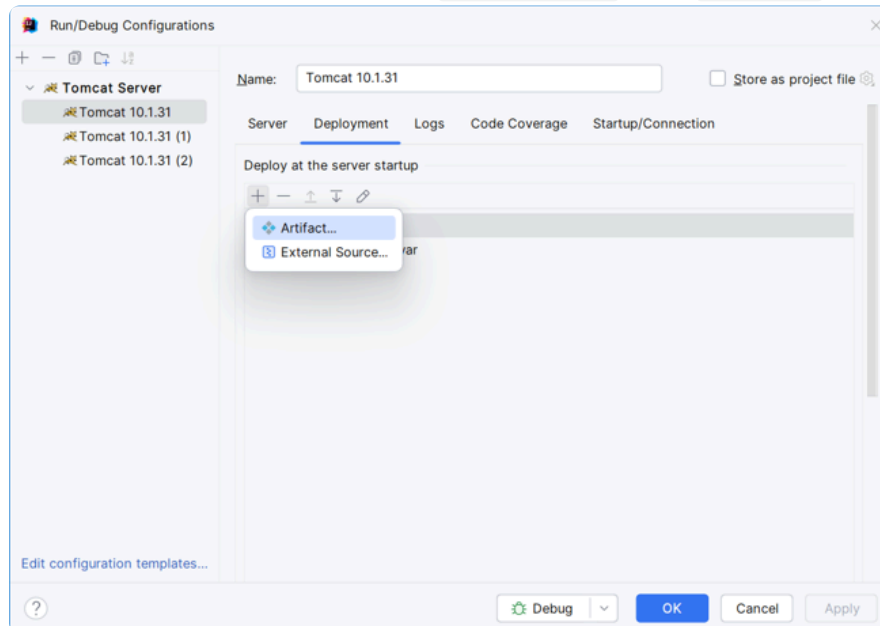
8. Download Tomcat 10.1.

9. Copy the jar files in `.m2/repositories/com/oracle/ojdbc7/12.1.10.1/ojdbc7-12.1.0.1.jar` and `.m2/repositories/com/oracle/ucp/11.2.0.4/ucp-11.2.0.4.jar` to your tomcat/lib folder.

a. If you are in PG, copy the `org/postgresql/postgresql/42.7.4/postgresql-42.7.4.jar` into the lib folder of your tomcat directory.

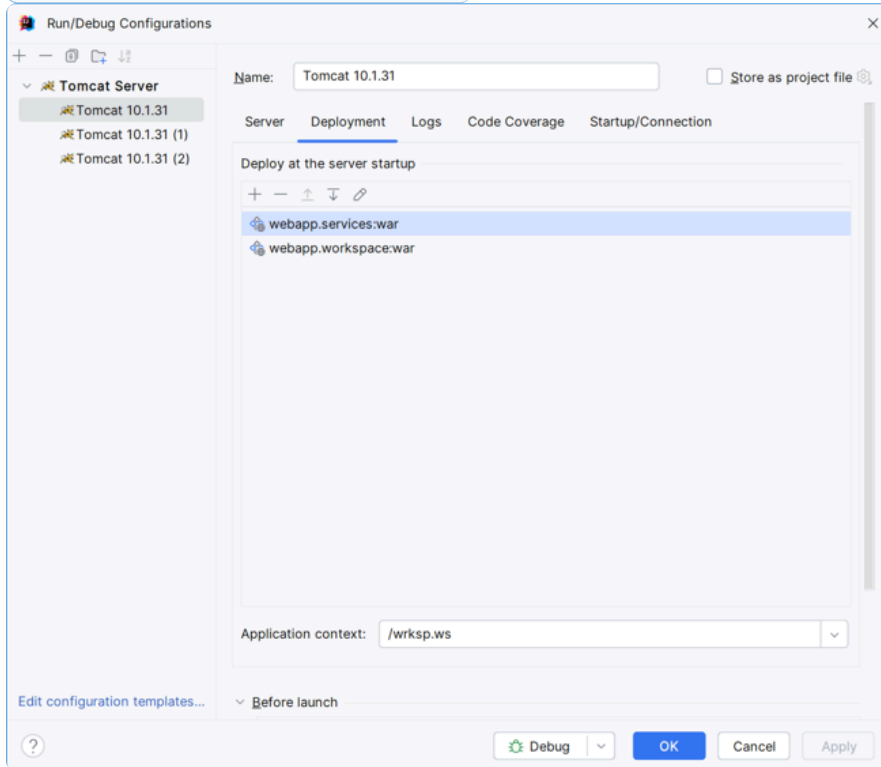
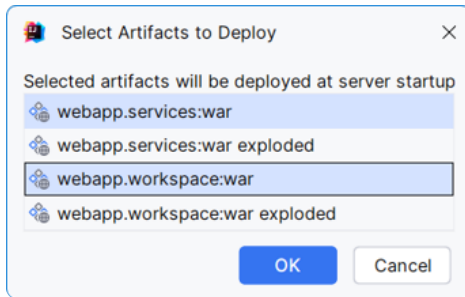
10. Create a new Tomcat configuration pointing to the downloaded Tomcat server.

a. Add webapp.services / API and webapp.workspace WAR files to the deployment tab, and set their deployment assembly to `/wiksp.ws` and `/wiksp`, respectively.

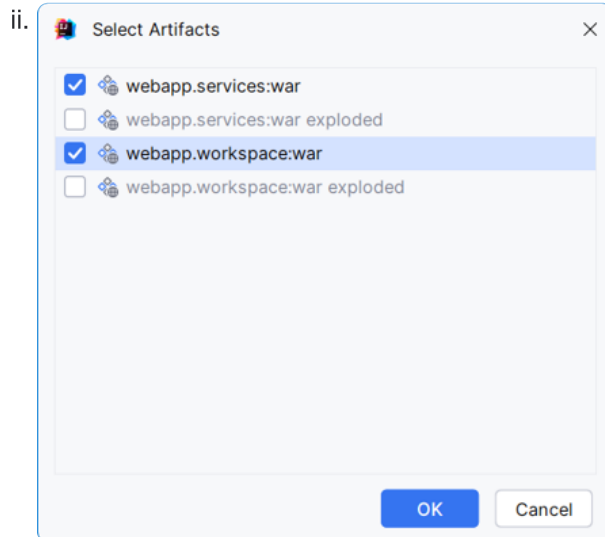
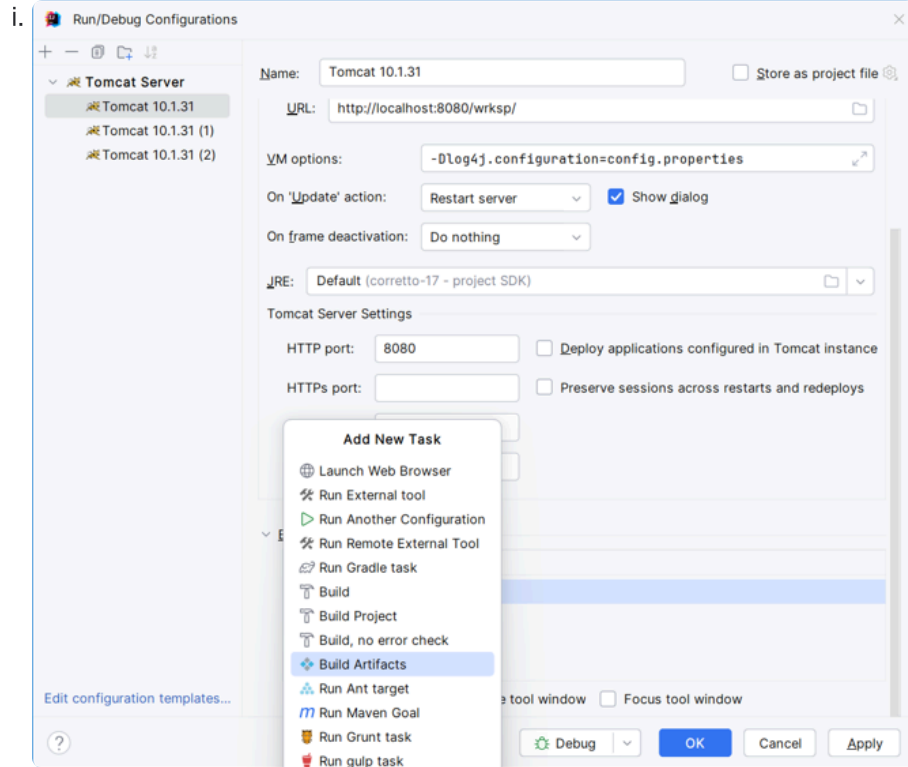


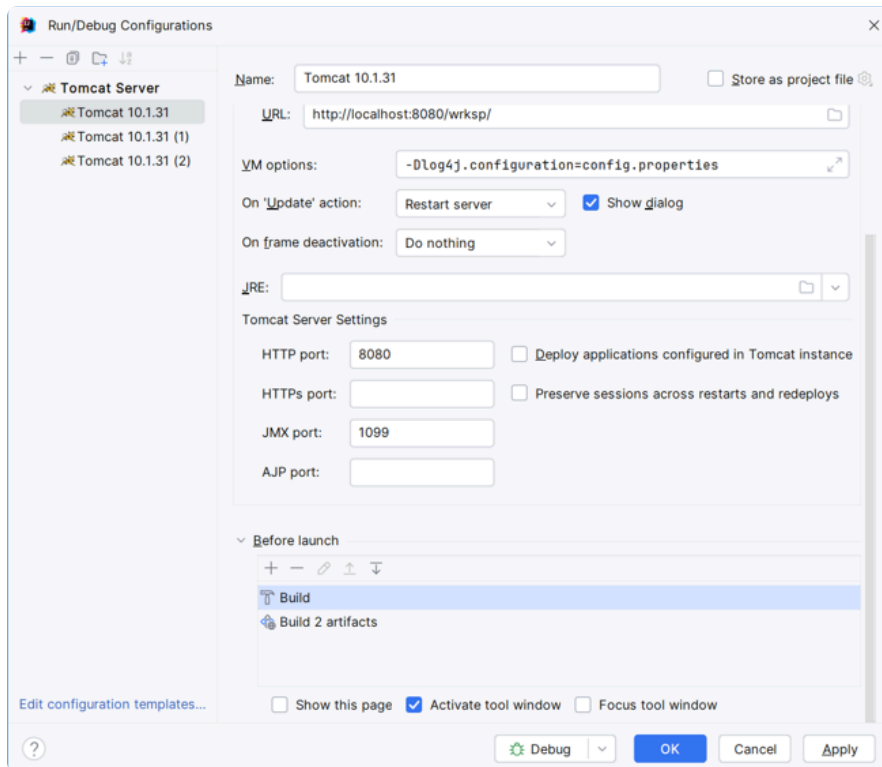
i.

ii.



- b. Set the configuration to build the 2 WAR files before launching it (you want the non-exploded WAR files), add `-Dlog4j.configuration=config.properties` to the VM options, set the URL to `http://localhost:8080/wrksp/` and set the JRE to Java 8 JRE

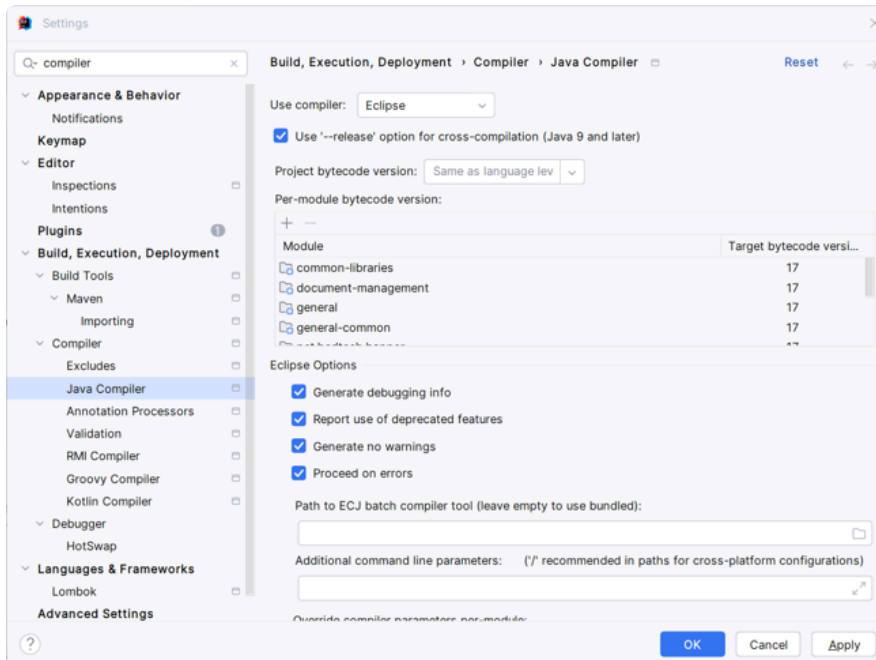




iii.

11. Go to Java settings and set the compiler from `javac` to `eclipse`. This will allow the compiler to ignore various runtime errors in the code, but IntelliJ will still highlight them.

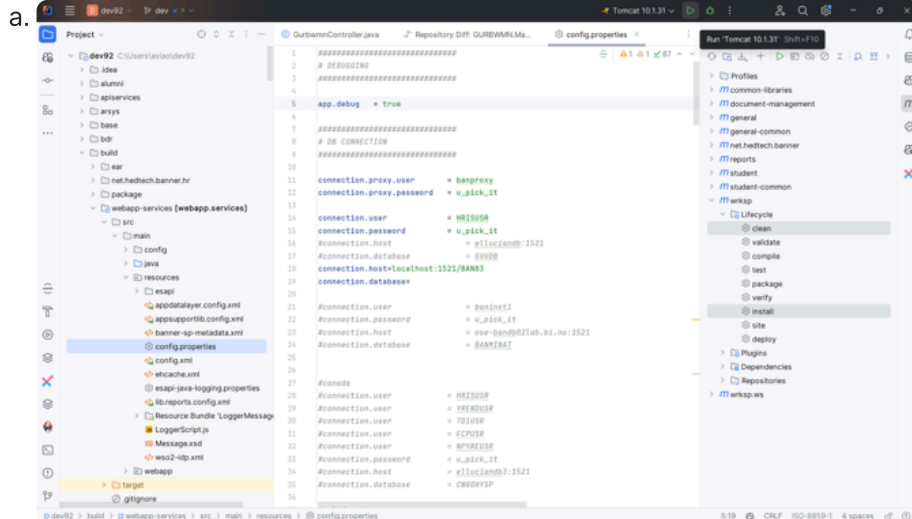
a.



b. You may find that the "Generate no Warnings" setting speeds up your startup time or decreases memory use

12. Go to `webapp-workspace` in the `workspace` repository and run `mvn clean install`. This will likely fail, but it appears to be needed to get some necessary files in your `webapp-workspace` target folder.

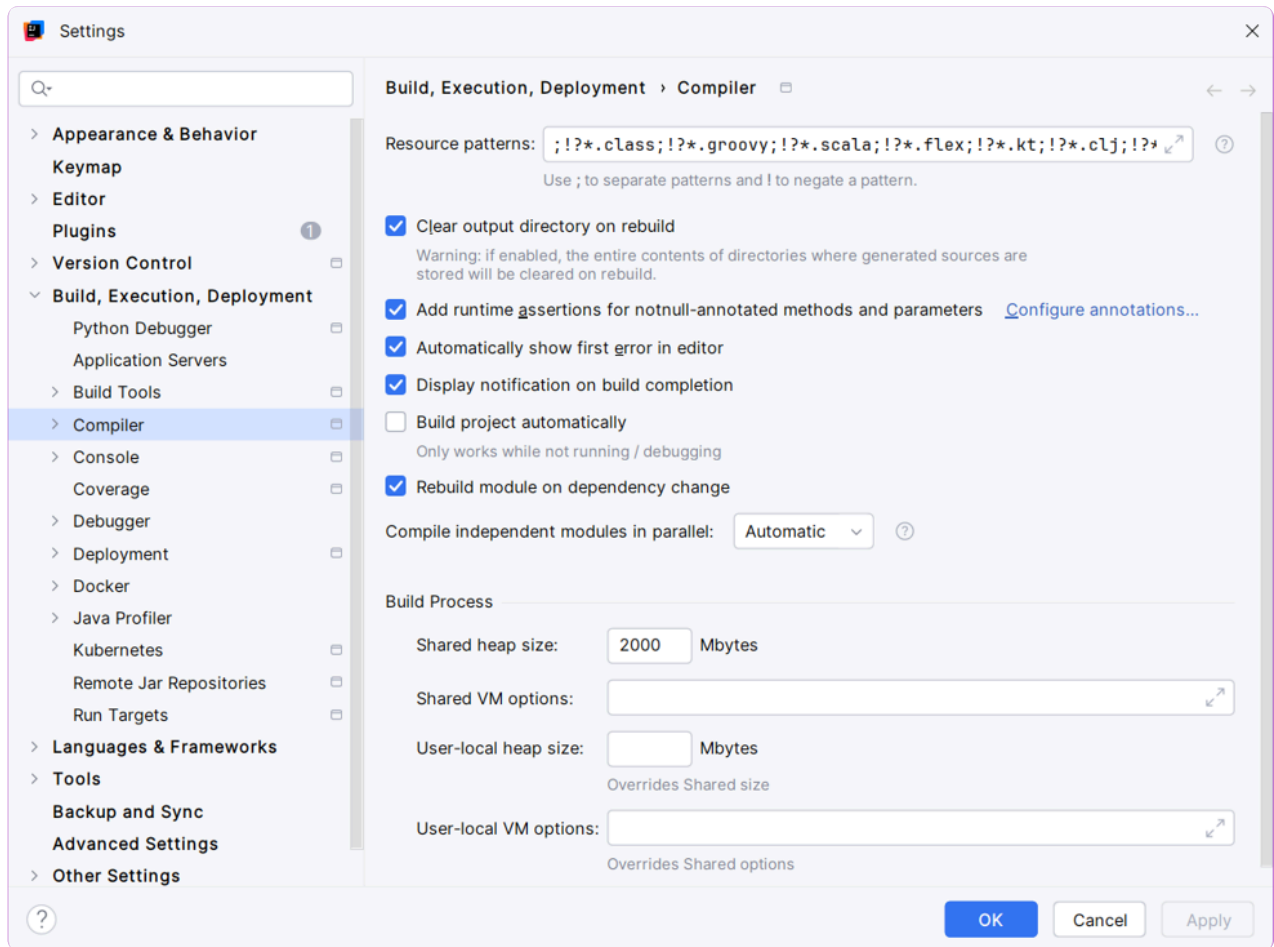
13. Go to `config.properties` in `webapp-services` and replace `connection.host` and `connection.database` settings (You can add use the IP addresses and database names of external databases in those fields, and you can have multiples of those fields to be commented or uncommented to rotate between them more easily.).
- Use `localhost:1521` for `connection.host` and `BAN83` for `connection.database` to use your local database in Oracle. If that does not work, use `localhost:1521/BAN83` for `connection.host` and leave `connection.database` empty
 - Use `localhost:5432` for `connection.host` and `ban2020` for `connection.database` to use your local database in PG.
 -
14. To run Banner Admin, click on the Run or Debug button at the top right corner of the UI, a window will automatically open once the server is launched



Common Issues

- If you see a blank page that only says `/wrksp.ws`, then change the homepage on the URL on the Server tab to `/wrksp`
- If you see yellow warning icons when launching Tomcat, remove the war files from the Deployment tab and readd them
- If you see red errors when launching Tomcat, fix your Tomcat configuration by pointing IntelliJ to the Tomcat installation folder in the run configuration, remove and readd the war files from the Deployment tab, and then restart IntelliJ (this seems to happen once every few days)
 - If you are still getting errors after this (IntelliJ opening a popup saying that the configuration is invalid even when there are no errors when you run the configuration), copy the configuration and run that one instead

- **Both Oracle and PG now use Java 17 + Tomcat 10, make sure to add the 2 Oracle jars and 1 PG jar to Tomcat!**
- In SaaS, the pom.xml file in build does not have other products commented out by default, so you have to do it yourself
- If the page opens and displays an error on the server not serving / refusing to serve the content, check your application context
- You may want to delete your entire project and re-clone your repos if you are migrating from pre-Spring 6 to Spring 6 (and Java 8 / 11 to Java 17)
- If code changes to Admin pages do not seem to be reflected on the server, clear the Tomcat cache (IntelliJ does not have a feature to do this like Eclipse) and delete the target folder of your repository (webapp.workspace or apiservices)
- If you cannot login, uncomment `<entry key="PKG_XML" value="goktxml"/>` in `appdata\layer.config.xml`
- If you are getting a yellow X while trying to start your tomcat server, clean the war file that has an error
- If you are getting issues with refreshing, try running `mvn clean install` from your command line
- IntelliJ may have issues modifying files in the Program Files directory, putting your Tomcat directory there may cause issues with launching
- Make sure to download the Windows version of the Tomcat if you are on Windows (the file inside the zip files have the same name)
- If you are having memory issues, increase the shared heap size setting in the compiler settings



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DBMS_OUTPUT in IntelliJ

In order to see DBMS_OUTPUT in IntelliJ, you need to click on the console section and then enable the DBMSOUTPUT button in your database console window

